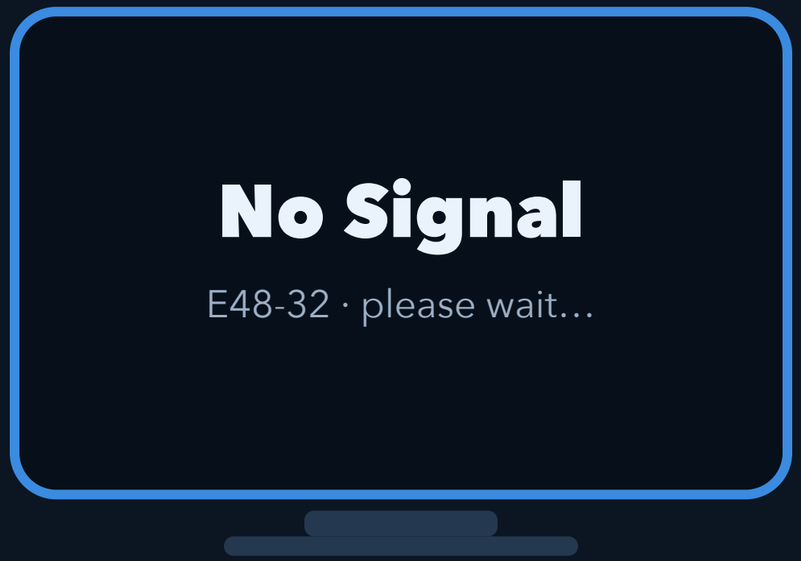
WEEK 1 · DAY 2

Charts That Don't Lie 📊

The same data can tell the truth or tell a lie.
Today you learn to draw the truth.

NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy



TODAY'S MISSION · 2 HOURS

Three moves



Catch the lie

the average hid every outage



Compare 6 zones

small multiples + index to week 1



Tell the story

a title + 4 sentences your boss reads

⏱️ 10 min hook

🎤 30 min slides

👩🏾‍💻 60 min hands-on

🏁 20 min wrap-up

BE HONEST

Who has seen this during heavy rain? 📺🌧️



That's **rain fade**. Ka-band suffers it most. Today's question:
did our Ka-band link actually go down in August, and would our charts show it?

PART 1

The average lies 🤥

One month of hourly Ka-band link data. One big trap.

TODAY'S FIRST DATASET · RF_LINK_QUALITY_HOURLY.CSV

August, hour by hour: 744 rows

🌧️

rain_rate_mmhr

rain at the earth station

📶

cn_ratio_db

signal quality (carrier-to-noise)

🔊

uplink_power_dbm

how loud we transmit

⛽

link_margin_db

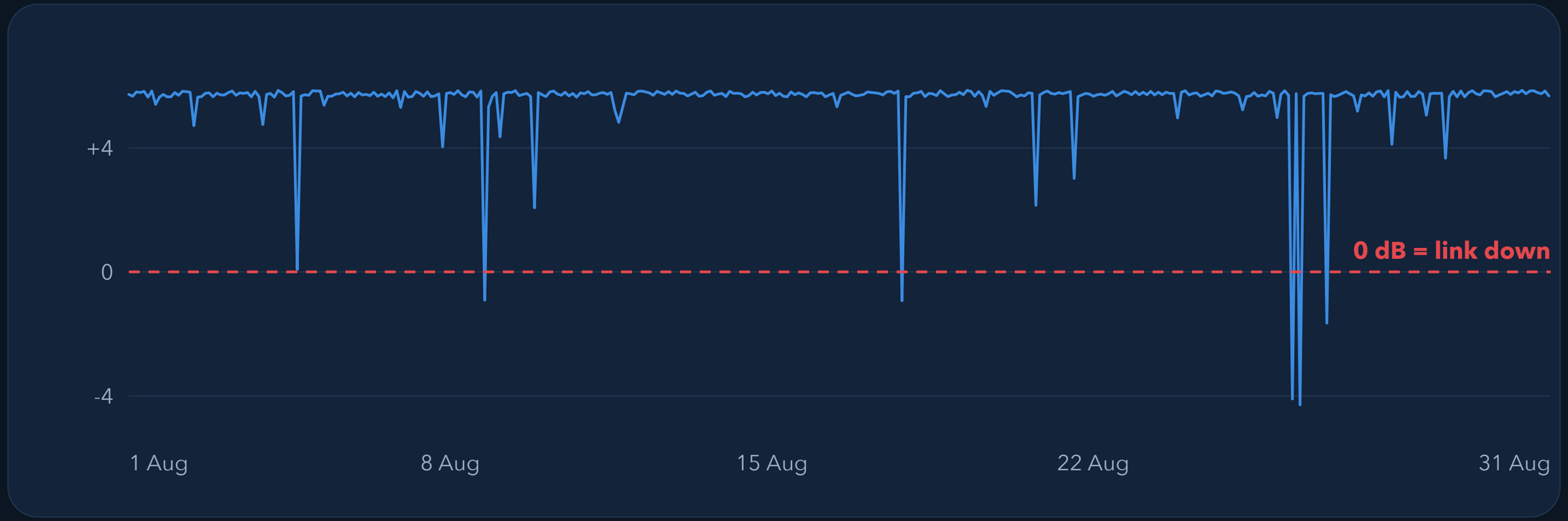
signal left above the minimum



Link margin is like fuel above the reserve line. Positive = you're fine. **Below 0 dB = the link is DOWN.**

EVERY HOUR OF AUGUST

Link margin dived below 0 dB 12 times



12 outage hours | **56 hours had rain** | **every outage came during heavy rain** | **worst hour: -4.29 dB**

SAME DATA. TWO DAILY SUMMARIES.

One says “all good”. One tells the truth.

Daily AVERAGE



Never below **4.91 dB**. Zero outages. 😎

Daily MINIMUM



11 of 31 days had an outage. 😱

THE RULE

“Did we breach?” → plot the worst moment

📉

Daily minimum

SLA, outages, “did it ever go down?”

⚖️

Daily average

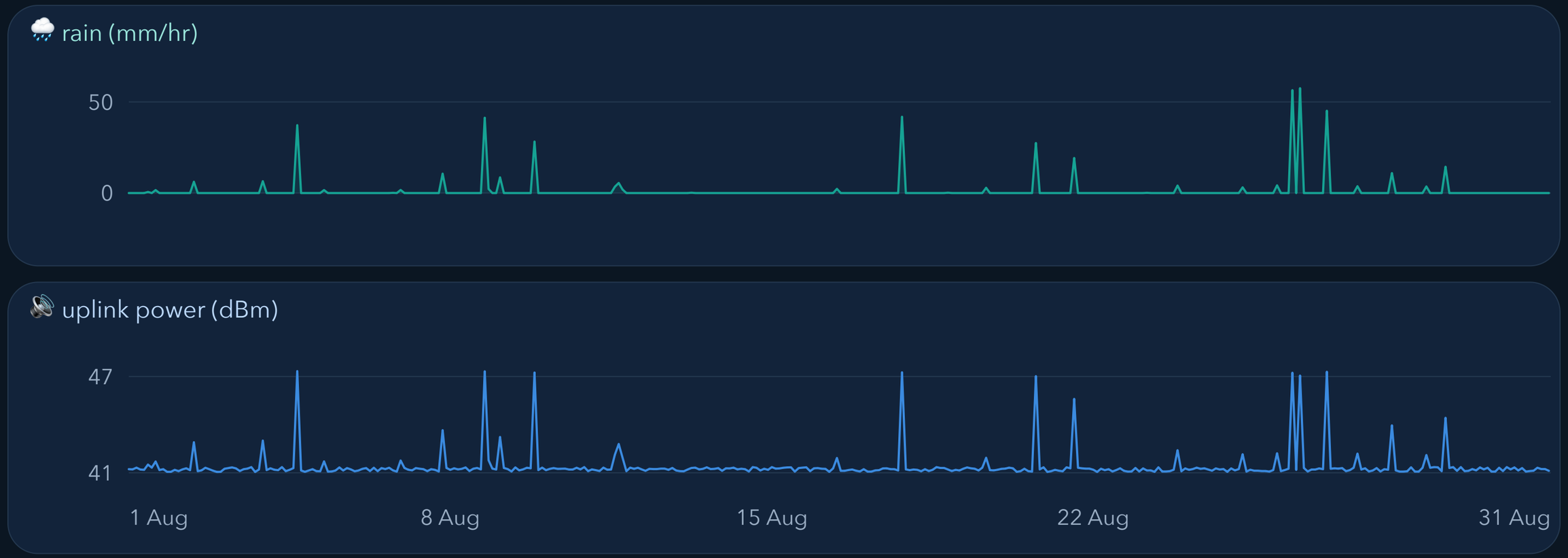
typical level, long-term trend



“Average traffic on Third Mainland Bridge was fine today.”
Tell that to the man who sat there for 3 hours. 😤

TWO MEASURES? TWO CHARTS, ONE TIME AXIS

Uplink power rises *because* it rains



Correlation **0.97**: the station turns up the volume to fight the rain. A **response**, not a second problem. Two units → **two stacked charts**, never two y-axes.

PART 2

Six zones, one picture 🗺️

VSAT uptime for Nigeria's six geopolitical zones, every week.

VSAT_UPTIME_BY_REGION_WEEKLY.CSV · 312 ROWS

Long format → pivot → wide format

LONG: one row per zone per week

week_start	region	uptime
2025-01-06	North West	99.02
2025-01-06	North East	98.33
2025-01-06	South West	99.62
2025-01-06	South East	99.14
2025-01-06	North Central	99.24
2025-01-06	South South	97.99

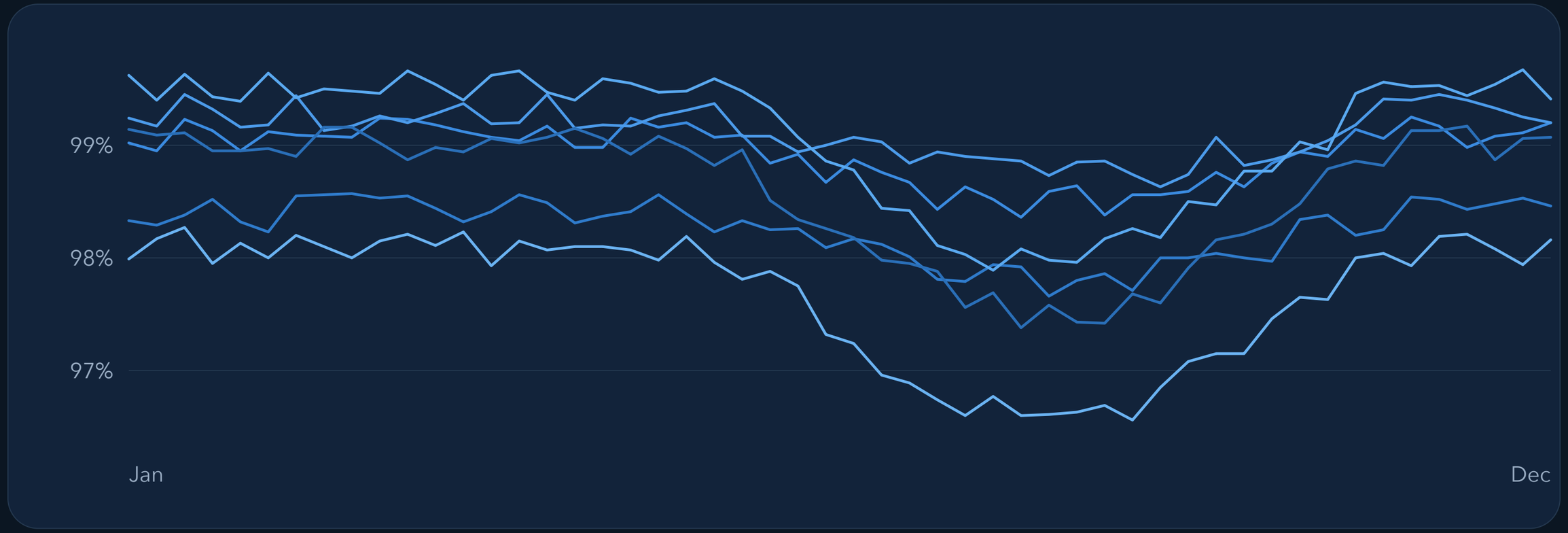
WIDE: one row per week, one column per zone

week	N. West	N. East	N. Central	S. West	S. East	S. South
2025-01-06	99.02	98.33	99.24	99.62	99.14	97.99
2025-01-13	98.95	98.29	99.17	99.40	99.09	98.17
2025-01-20	99.23	98.38	99.45	99.63	99.11	98.27

```
wide = vsat.pivot(index="week_start", columns="region", values="uptime_pct")
```

OPTION A

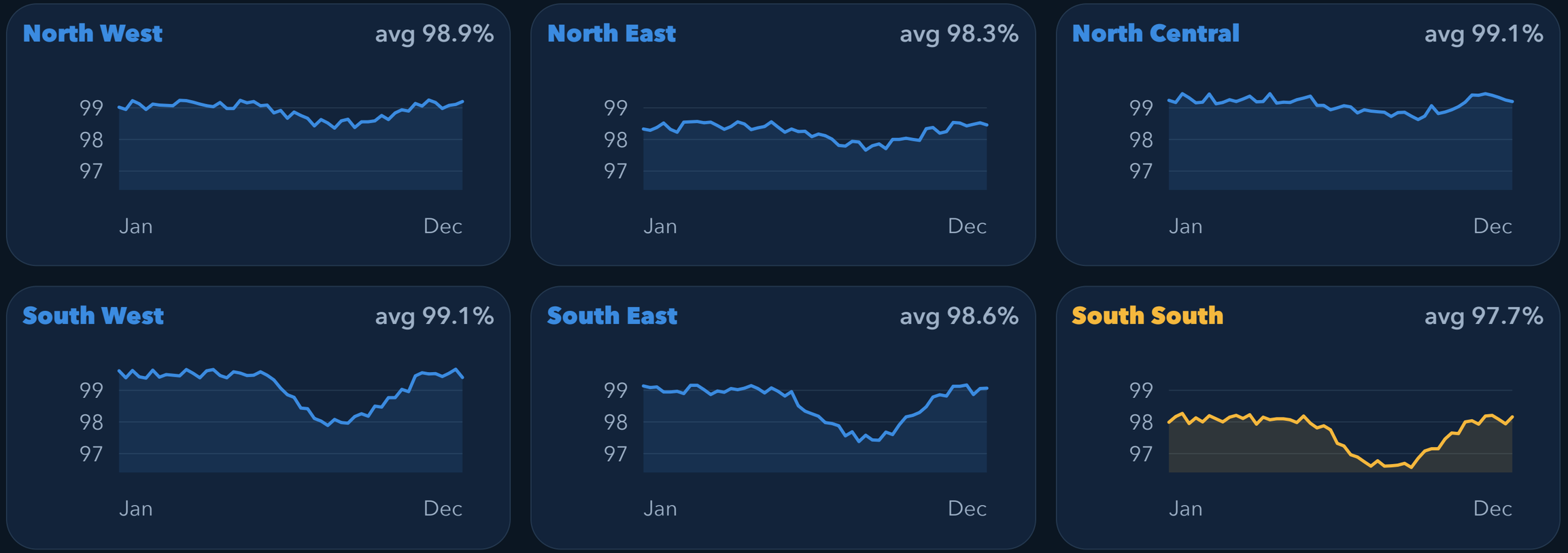
Six lines on one chart. Find South South. 🍝



Spaghetti! Colours fight, lines cross, nobody can read it. 😵‍💫

OPTION B · SMALL MULTIPLES

One mini chart per zone, same y-axis



Instantly clear: **South South** sits lowest all year (avg 97.67%), and the 3 southern zones dip harder in the rainy season.

A DIFFERENT QUESTION

Who fell furthest from their own normal?

```
indexed = wide / wide.iloc[0] * 100      # every zone starts at 100
```



South East · South West · South South · the 3 northern zones

TWO QUESTIONS, TWO DIFFERENT ANSWERS

The chart you draw depends on the question



“Who is worst?”

raw uptime, small multiples

South South: lowest average (97.67%) and the worst single week (96.56%).



“Who dropped most from normal?”

indexed to week 1

South East: fell to 98.2 (week 1 = 100), just ahead of South West.

Like comparing students: **lowest score** is not the same as **biggest drop since last term**. 📚

PART 3

Honest-chart rules ✅

Five habits that make your charts trustworthy.

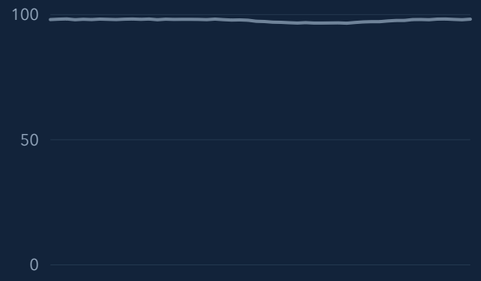
WHERE SHOULD THE Y-AXIS START?

It depends on what you measure

📈 Percentages near the top

South South uptime

0–100: flat, useless



96–99: story visible ✅

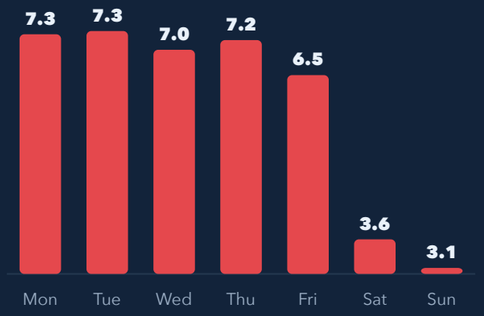


Zoom in, and **write the range on the chart**.

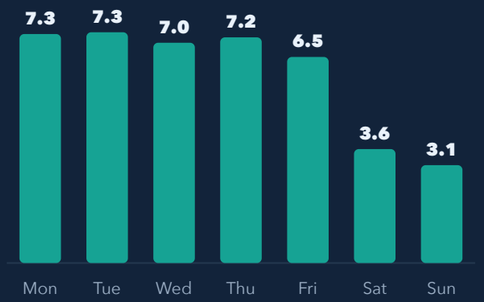
🎟️ Counts (tickets)

average NOC tickets per weekday

axis from 3 ❌ Sunday looks dead



axis from 0 ✅ honest



Things you can add up: **start at zero**.

BEFORE YOU SHARE ANY CHART

The 5 honest-chart rules



Threshold line

draw the limit (0 dB, SLA %)



Title = finding

say what it SHOWS



Sensible axis

and say the range



One y-axis

two units? two charts



Units

dB, %, tickets…

Think of your chart as **evidence in court**. The other lawyer will check all five. ⚖️

UPGRADE YOUR TITLE

A title should state the finding

❌ Names the data

Link margin, August

✅ States the finding (with a number)

Ka-band link went down on 11 of 31 August days, all during heavy rain

THE 4-SENTENCE FINDING

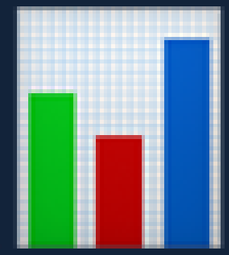
How to report it to your unit head

1. **Situation:** The Ka-band Nigeria beam carries customer traffic through the rainy season.
2. **Observation:** In August the link was down for 12 hours across 11 days, yet the daily average never fell below 4.91 dB.
3. **Implication:** Reports built on daily averages would show zero outages and miss every SLA breach.
4. **Recommendation:** Report availability from hourly minimums, with the 0 dB line on every chart.

YOUR TURN · 60 MINUTES

Open the W1D2 exercise 👩🏾‍💻

1. **Open in Colab → Save a copy in Drive** 💾
2. Run **STEP 0**, then top to bottom
3. Part A: catch the average in its lie
4. Part B: six zones, one honest picture



14 checks to pass

Part A: the average lies
Part B: six zones
🌶️ Bonus: what rain kills the link?

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



Breach questions → **plot the minimum.** The average hid 12 outage hours.



Many groups → **small multiples** on one shared axis. **Change from normal** → **index** to week 1.



Tell it: a title that states the finding + Situation → Observation → Implication → **Recommendation**.

Next time 👉 **Calm Down the Noise**: why even faults rest on Sunday 😴